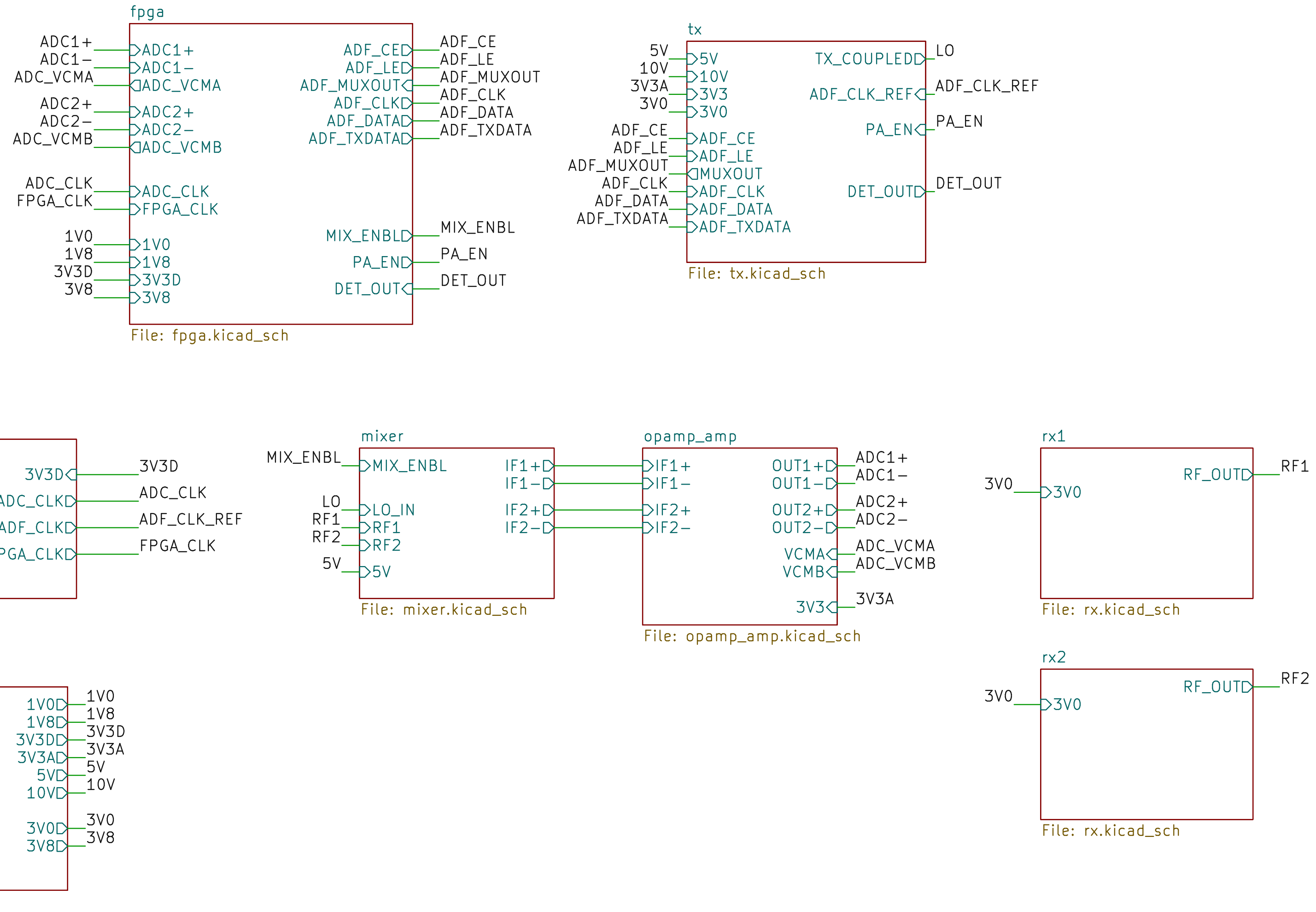
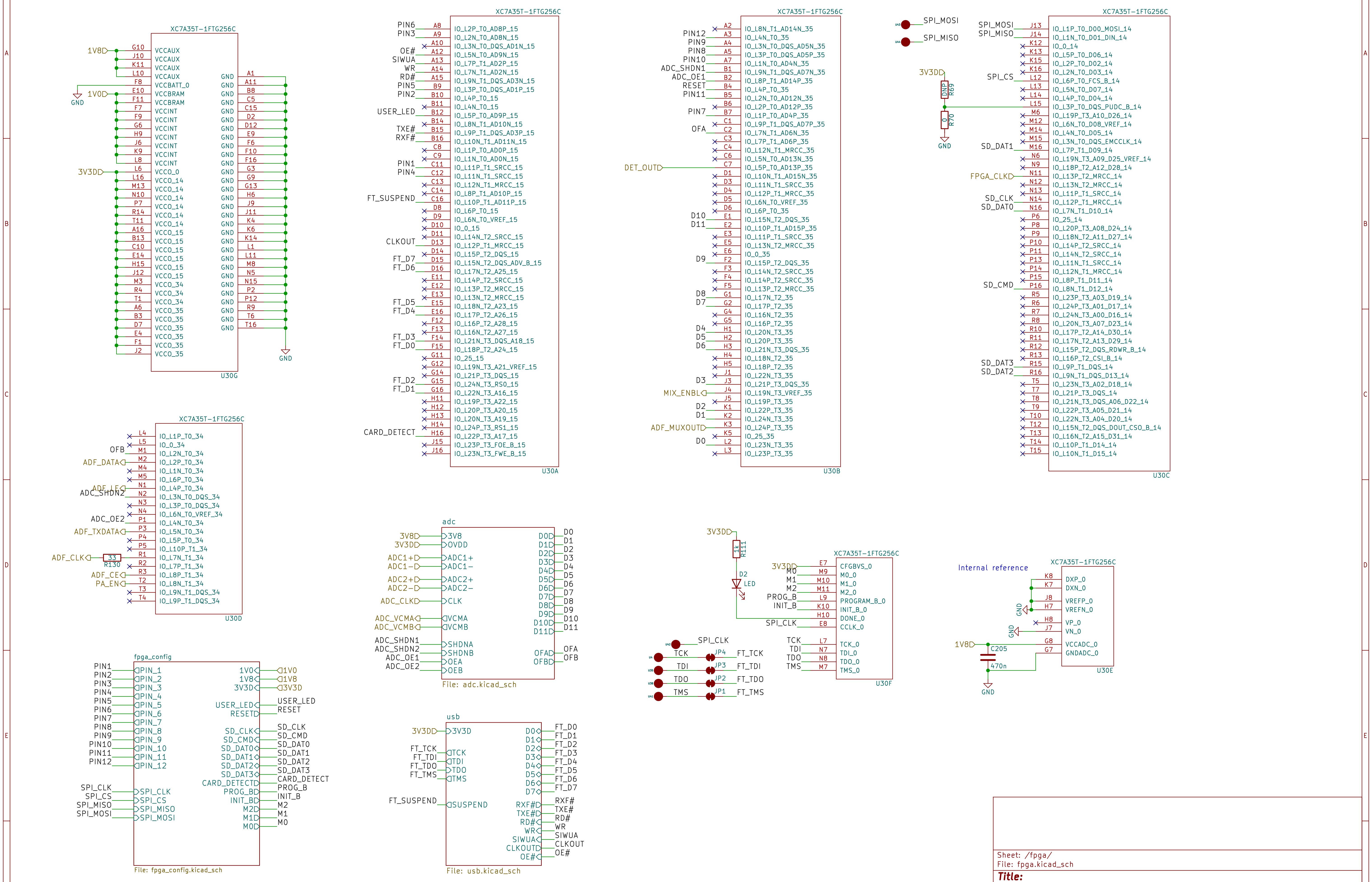



Sheet: /
File: fmcw3.kicad_sch

**Title: FMCW3 by CENK KESKİN**

| | | |
|---|---|---|
| Size: A4 | Date: | Rev: |
| KiCad E.D.A. 9.0.7 | | Id: 1/12 |

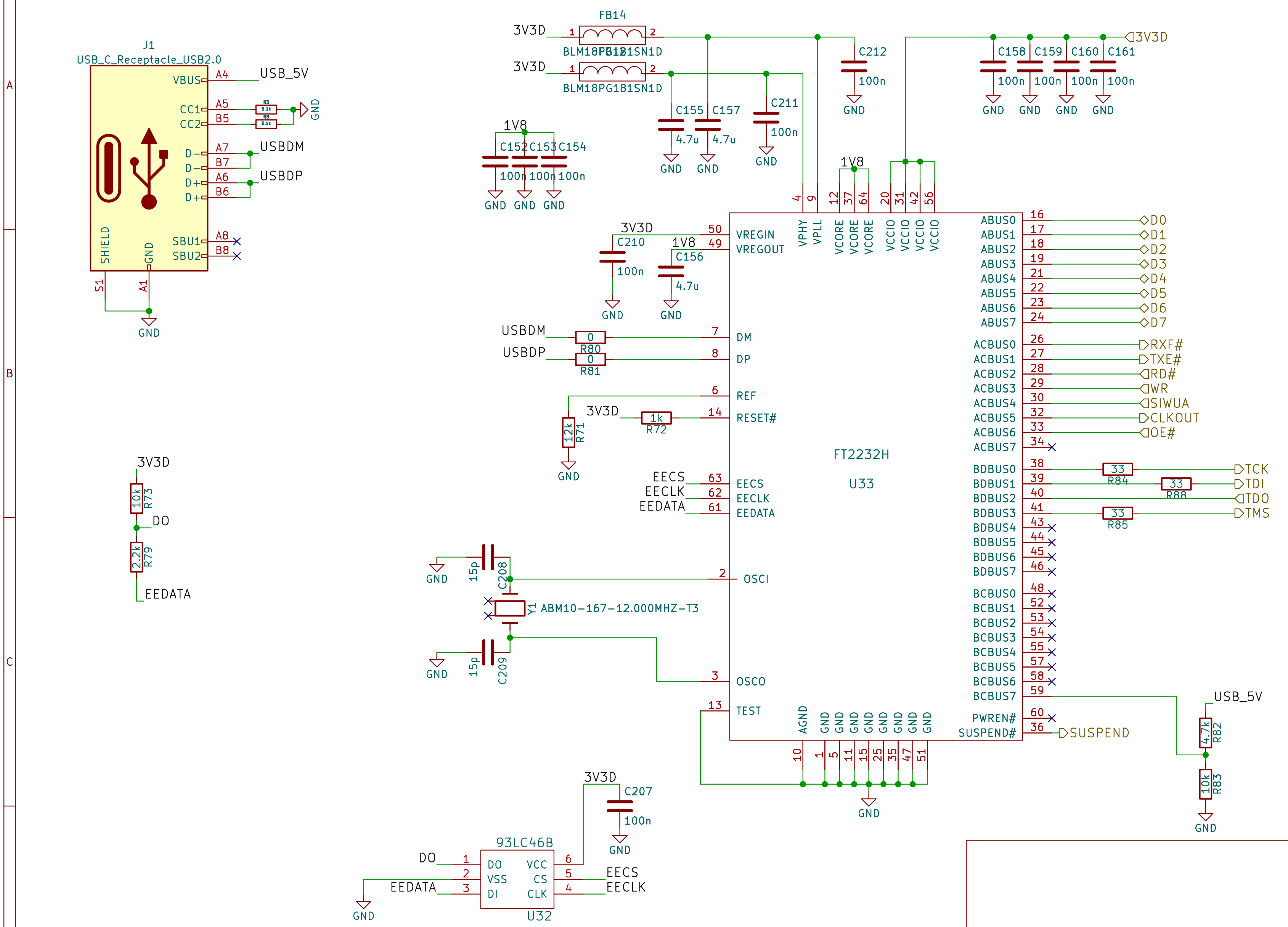



Sheet: /fpga/
File: fpga.kicad_sch

**Title:**

Size: A3 | Date: | Rev:

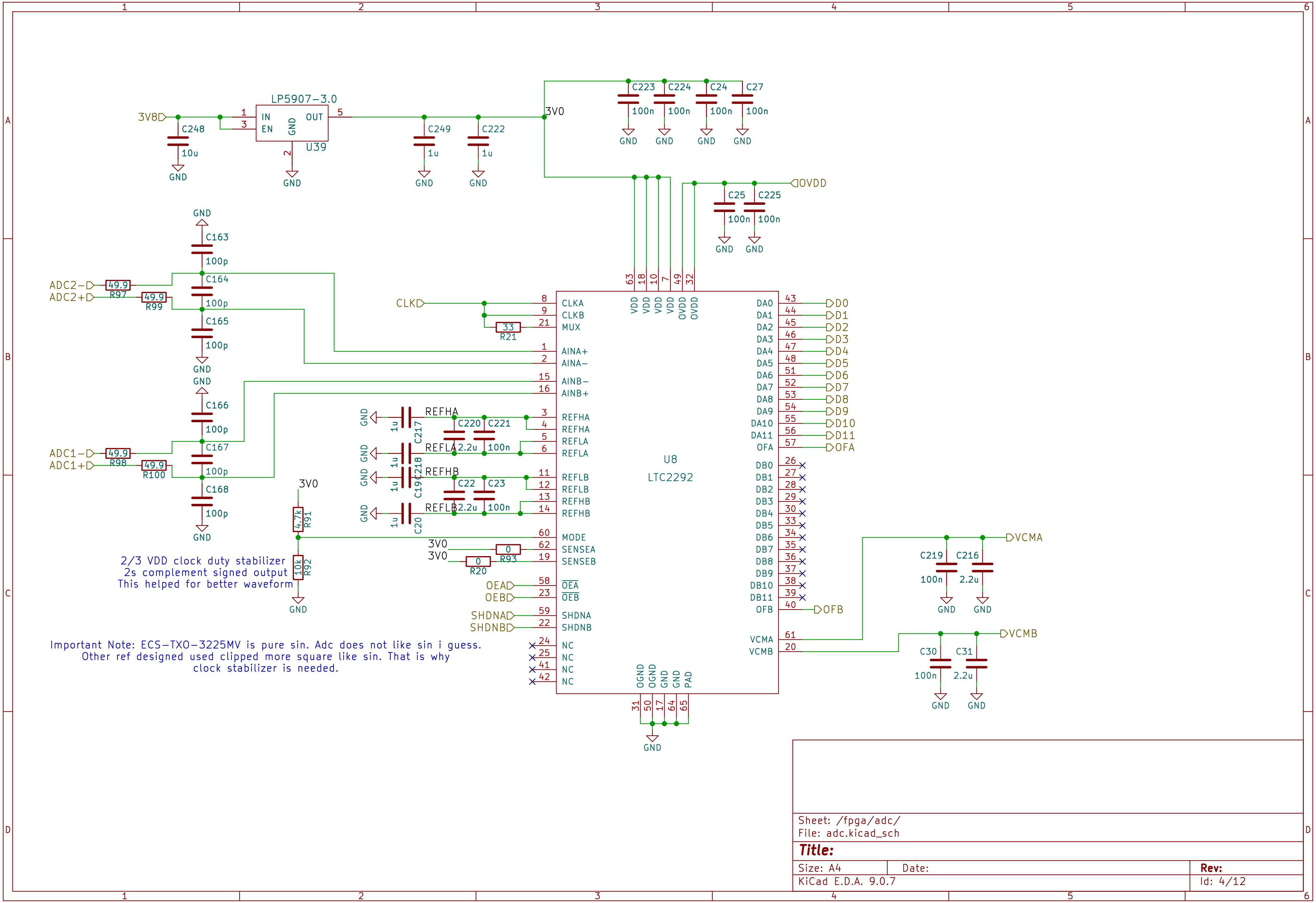
KiCad E.D.A. 9.0.7 | Id: 2/12



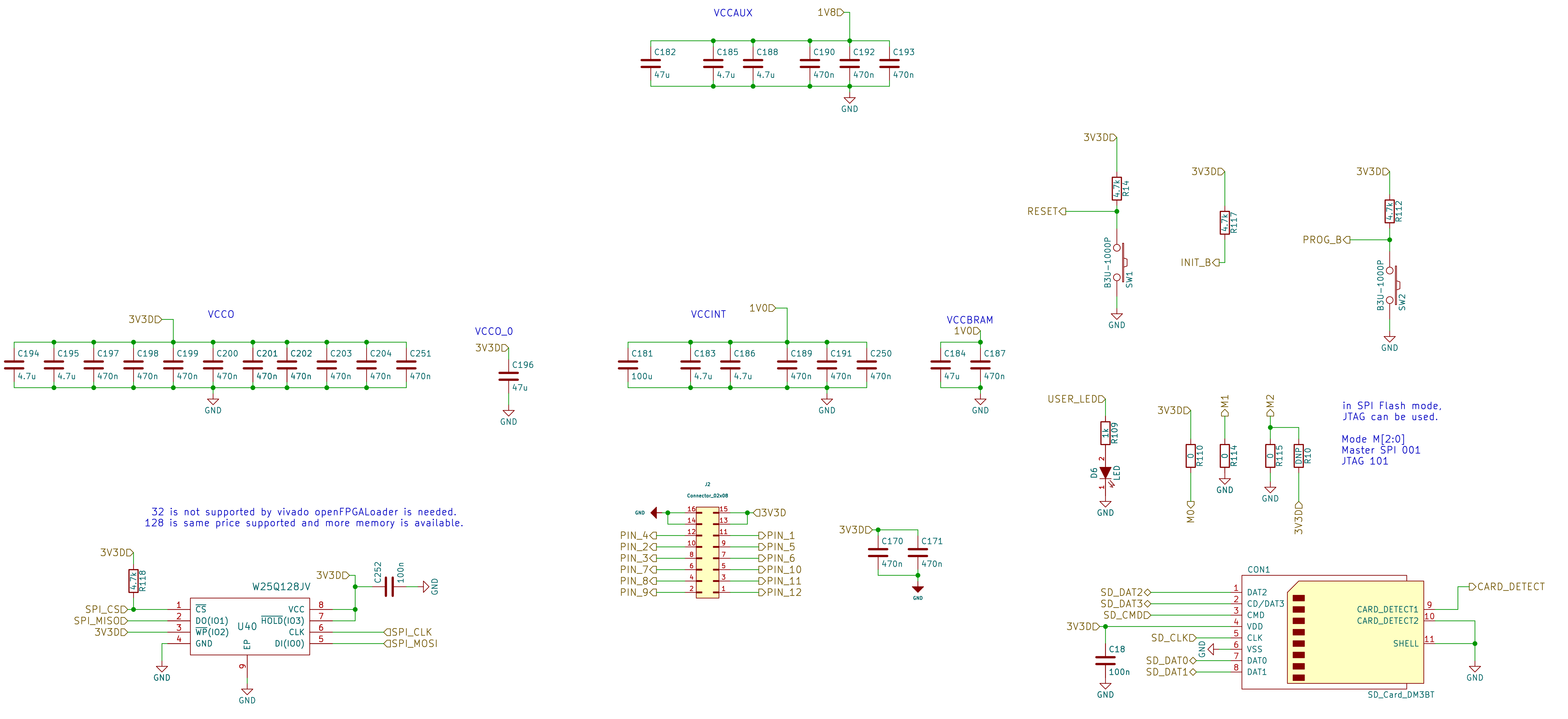

Sheet: /fpga/usb/
File: usb.kicad_sch

**Title:**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 9.0.7 | | Id: 3/12 |



LP5907-3.0
U39
IN
EN
OUT
GND
3V8
3V0
OVDD
CLK
ADC2-
ADC2+
ADC1-
ADC1+
U8
LTC2292
CLKA
CLKB
MUX
AINA+
AINA-
AINB-
AINB+
REFHA
REFLA
REFLB
REFHB
MODE
SENSEA
SENSEB
OEA
OEB
SHDNA
SHDNB
NC
VDD
OVDD
OGND
GND
PAD
VCMA
VCMB
OFA
OFB
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
2/3 VDD clock duty stabilizer
2s complement signed output
This helped for better waveform
Important Note: ECS-TXO-3225MV is pure sin. Adc does not like sin i guess.
Other ref designed used clipped more square like sin. That is why
clock stabilizer is needed.
Sheet: /fpga/adc/
File: adc.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 4/12

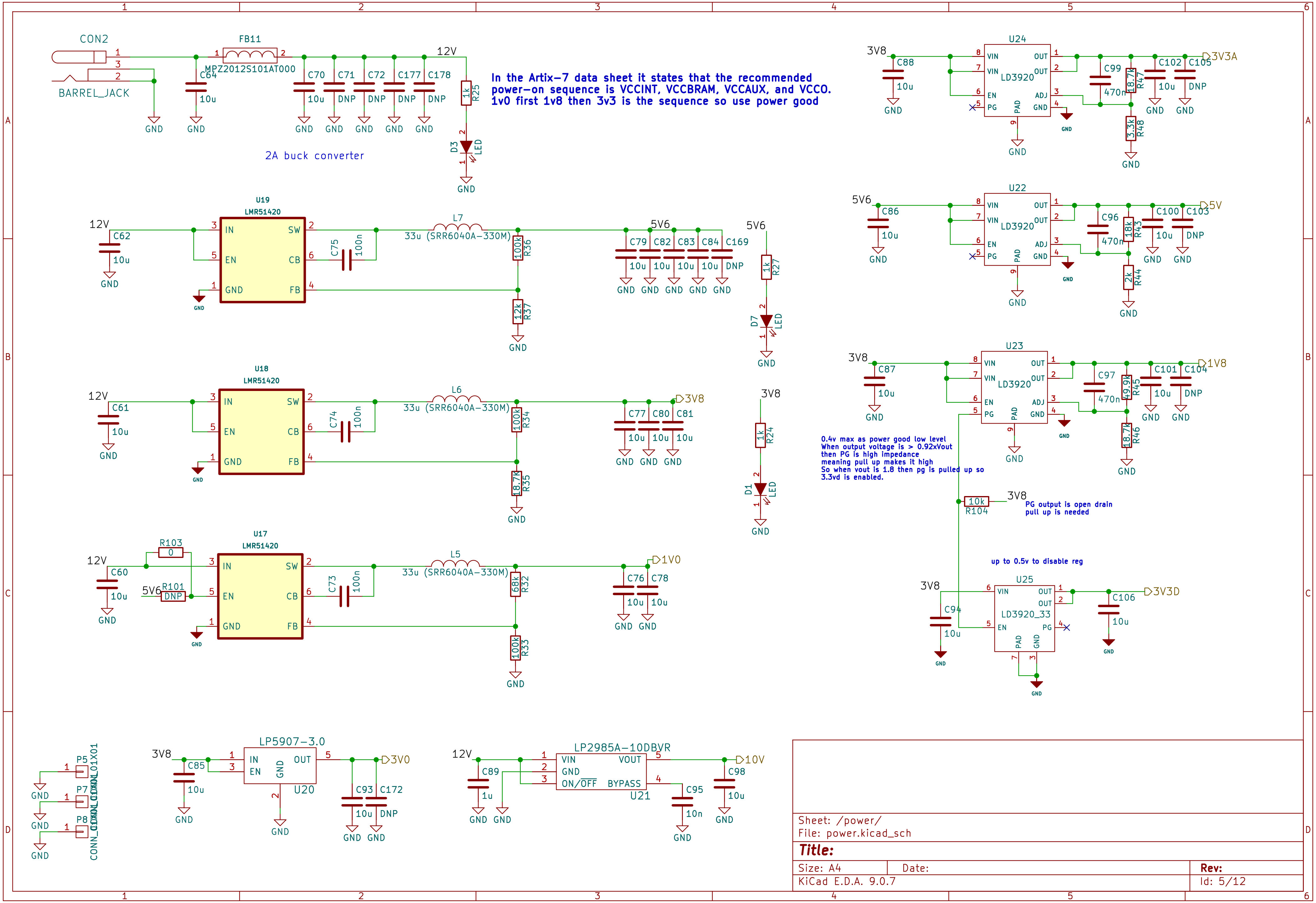



Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

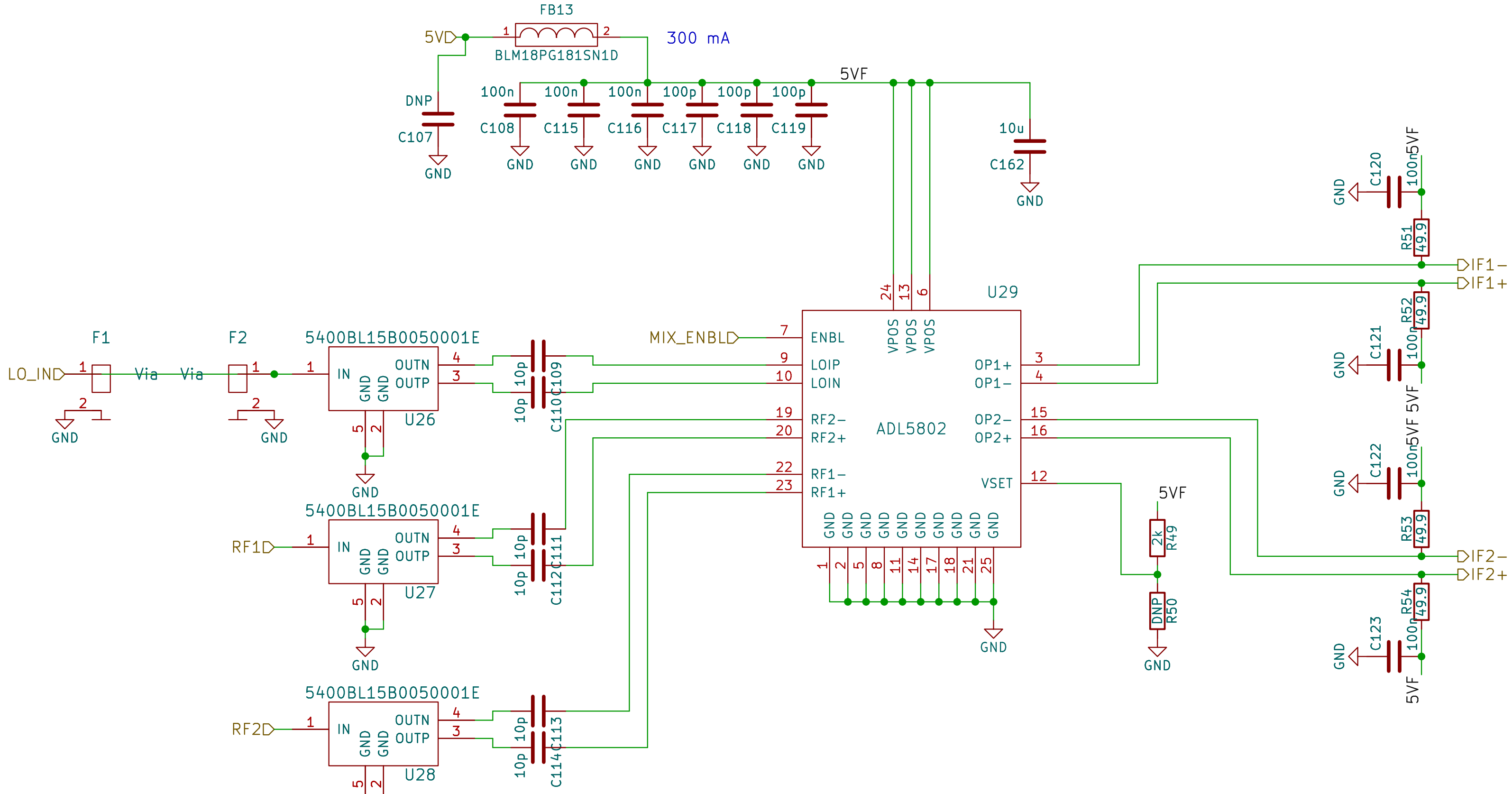
**Title:**

Size: A3 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 12/12



CON2
BARREL_JACK
FB11
MPZ2012S101AT000
C64 10u
C70 10u
C71 DNP
C72 DNP
C177 DNP
C178 DNP
12V
R25 1k
D3 LED
GND
2A buck converter
In the Artix-7 data sheet it states that the recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO. 1v0 first 1v8 then 3v3 is the sequence so use power good
U19
LMR51420
IN
EN
GND
SW
CB
FB
C62 10u
C75 100n
L7
33u (SRR6040A-330M)
R36 100k
R37 12k
5V6
C79 C82 C83 C84 C169
10u 10u 10u 10u DNP
R27 1k
D7 LED
U18
LMR51420
C61 10u
C74 100n
L6
33u (SRR6040A-330M)
R34 100k
R35 18.7k
3V8
C77 C80 C81
10u 10u 10u
R24 1k
D1 LED
U17
LMR51420
R103 0
R101 DNP
C60 10u
C73 100n
L5
33u (SRR6040A-330M)
R32 68k
R33 100k
1V0
C76 C78
10u 10u
U24
LD3920
VIN OUT EN PG PAD ADJ GND
C88 10u
C99 470n
R47 18.7k
R48 3.3k
C102 10u
C105 DNP
3V3A
U22
LD3920
5V6
C86 10u
C96 470n
R43 18k
R44 2k
C100 10u
C103 DNP
5V
U23
LD3920
C87 10u
C97 470n
R45 49.9k
R46 18.7k
C101 10u
C104 DNP
1V8
0.4v max as power good low level
When output voltage is > 0.92xVout
then PG is high impedance
meaning pull up makes it high
So when vout is 1.8 then pg is pulled up so
3.3vd is enabled.
R104 10k
PG output is open drain
pull up is needed
up to 0.5v to disable reg
U25
LD3920_33
C94 10u
C106 10u
3V3D
P5
P7
P8
CONN_01X01
LP5907-3.0
IN EN GND OUT
U20
C85 10u
C93 10u
C172 DNP
3V0
LP2985A-10DBVR
VIN GND ON/OFF VOUT BYPASS
U21
C89 1u
C95 10n
C98 10u
10V
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
KiCad E.D.A. 9.0.7
Rev:
Id: 5/12

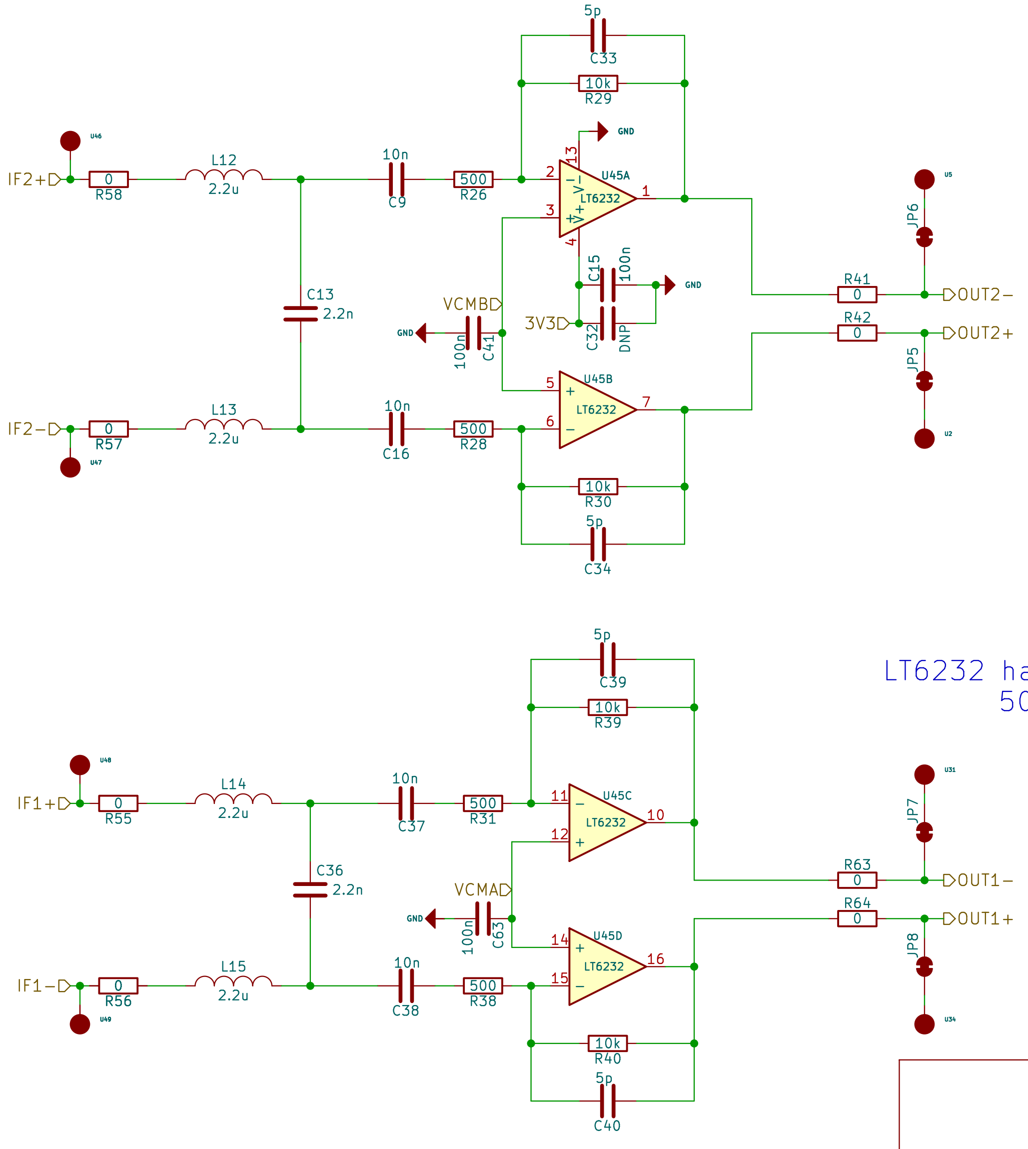



IMPORTANT: BALUN is 50:50 so output is differential 50 ohm.
Also ADL5802 expects 50 ohm differential as well.
Therefore rf trace width spacing changes at that part!!!

Sheet: /mixer/
File: mixer.kicad_sch

**Title:**

| | | |
|---|---|---|
| Size: A4 | Date: | Rev: |
| KiCad E.D.A. 9.0.7 | | Id: 6/12 |




FPGA FIR Filter works perfectly.
Second stage for HPF
range compensation filer is removed

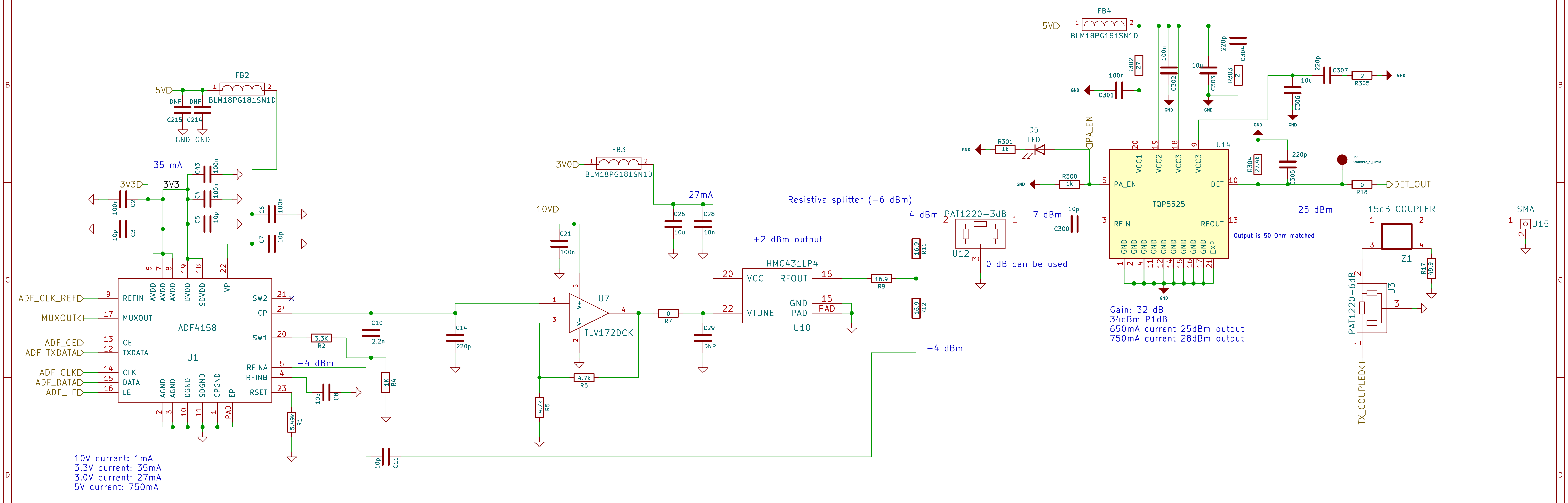
LT6232 has better noise performance as 1.1nV/√Hz
500 10K has less noise contribution

Sheet: /opamp_amp/
File: opamp_amp.kicad_sch

**Title:**

Size: A4 Date:
KiCad E.D.A. 9.0.7

**Rev:**
Id: 7/12

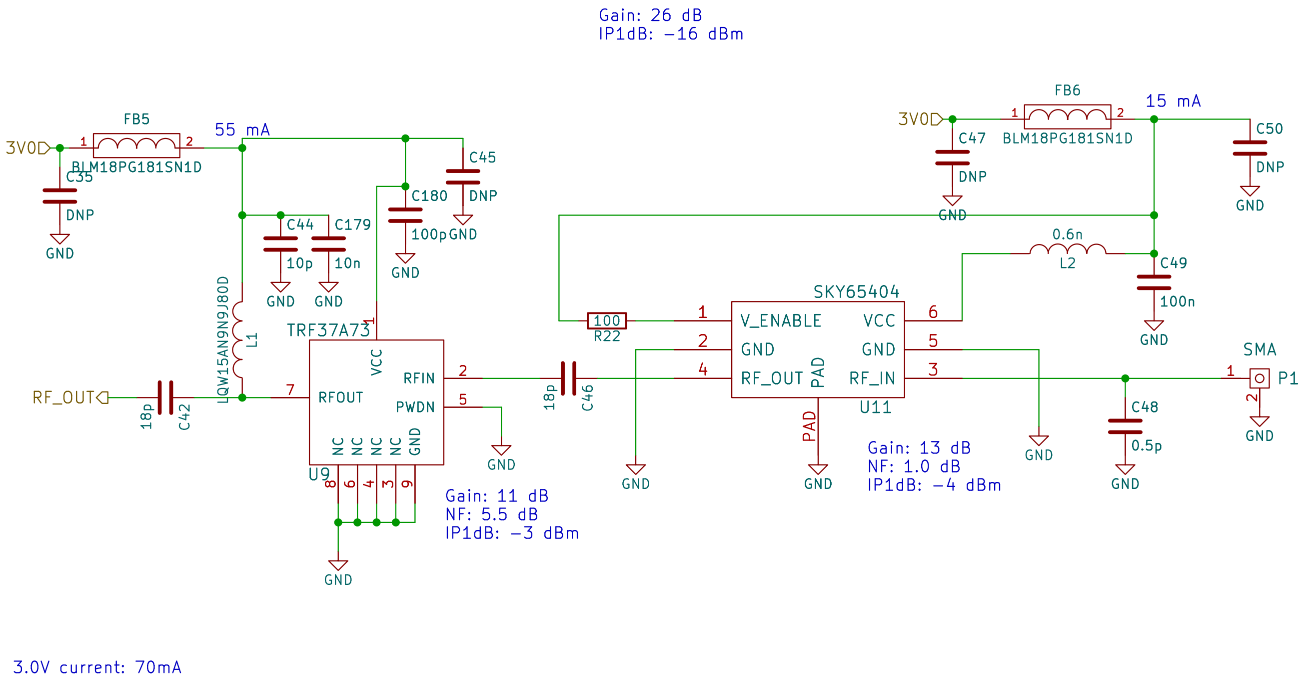



Sheet: /tx/
File: tx.kicad_sch

**Title:**

Size: A3 Date: Rev:

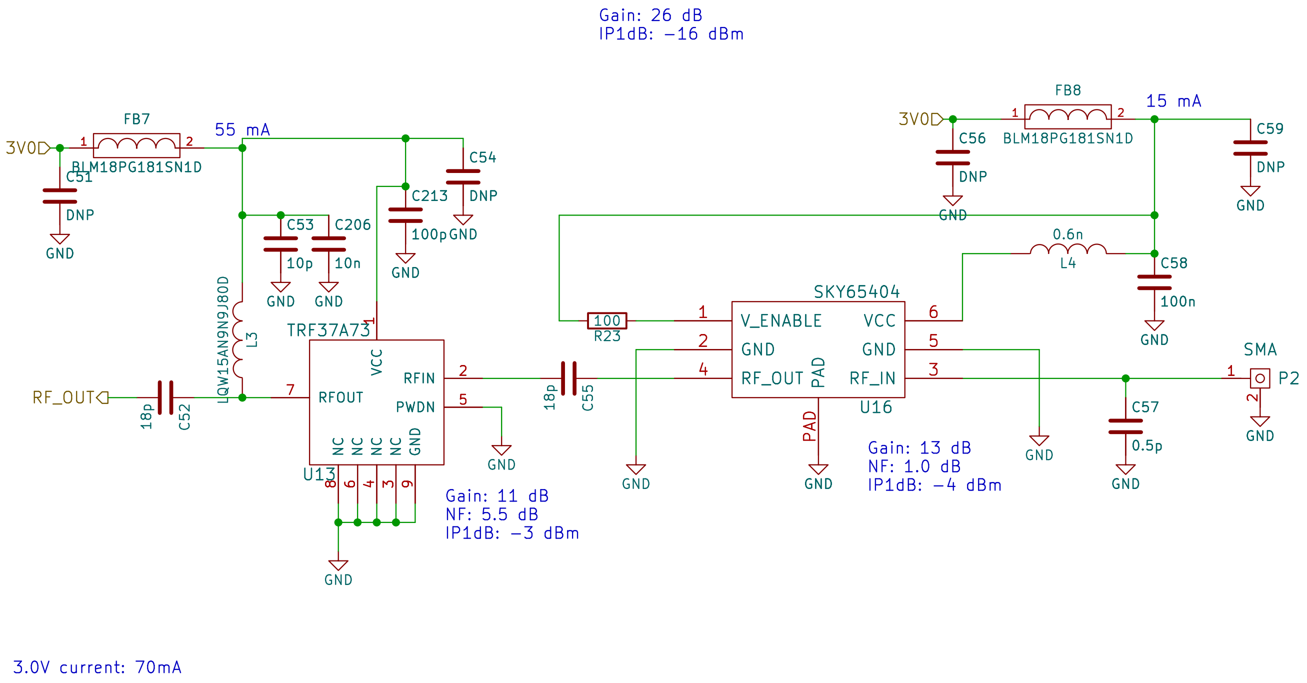
KiCad E.D.A. 9.0.7 Id: 8/12




Sheet: /rx1/
File: rx.kicad_sch

**Title:**

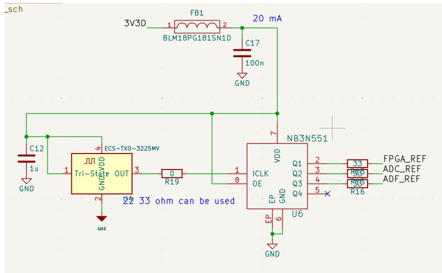
Size: A4
Date:
**Rev:**
KiCad E.D.A. 9.0.7
Id: 9/12



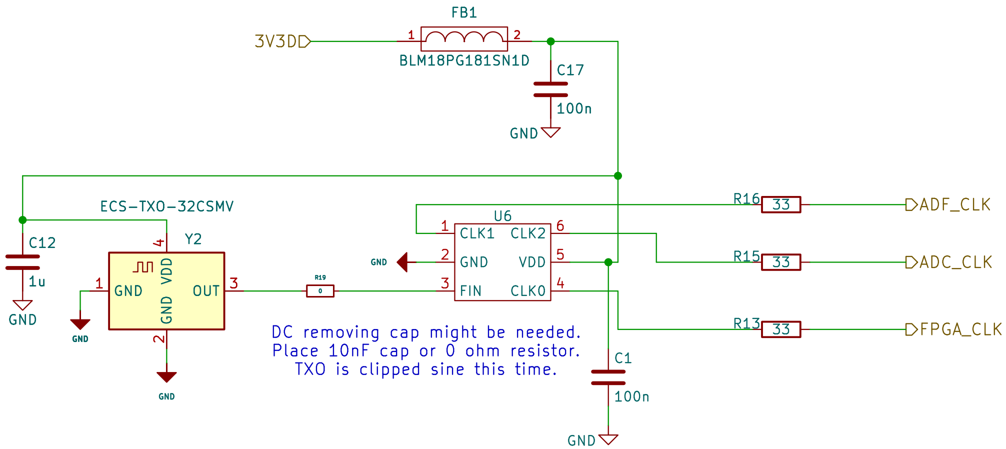

Sheet: /rx2/
File: rx.kicad_sch

**Title:**

| | | |
|---|---|---|
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.7 | | Id: 10/12 |




This is the prev design:
NB3N551 is obs.
Also this txo is generating pure sin.
ADC might need cmos like input not sin.
That could be the reason of duty stabilizer.




Sheet: /clock/
File: clock.kicad_sch

**Title:**

| | | |
|---|---|---|
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.7 | | Id: 11/12 |